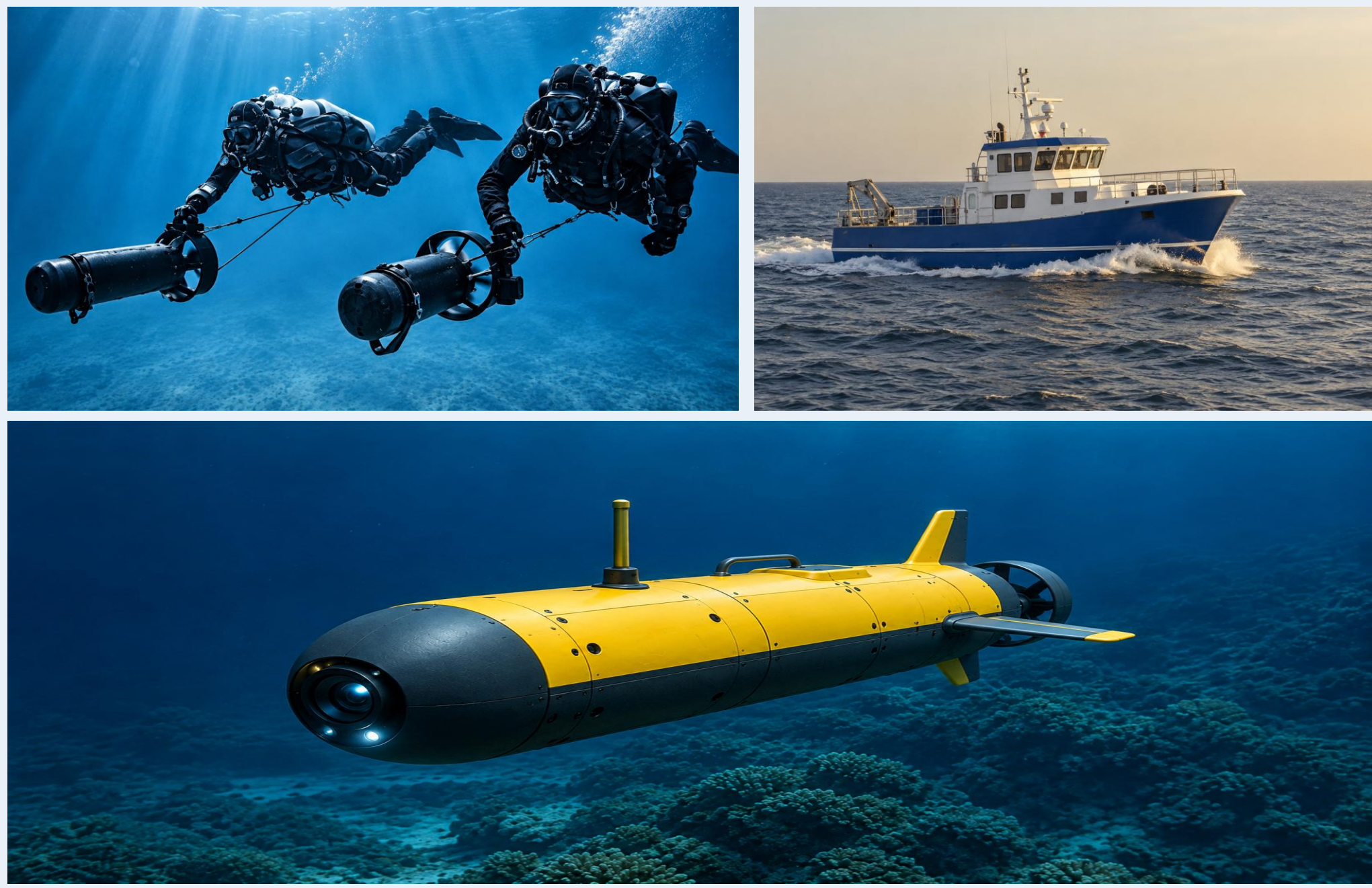


Physics-Informed Machine Learning Vessel Depth Classification

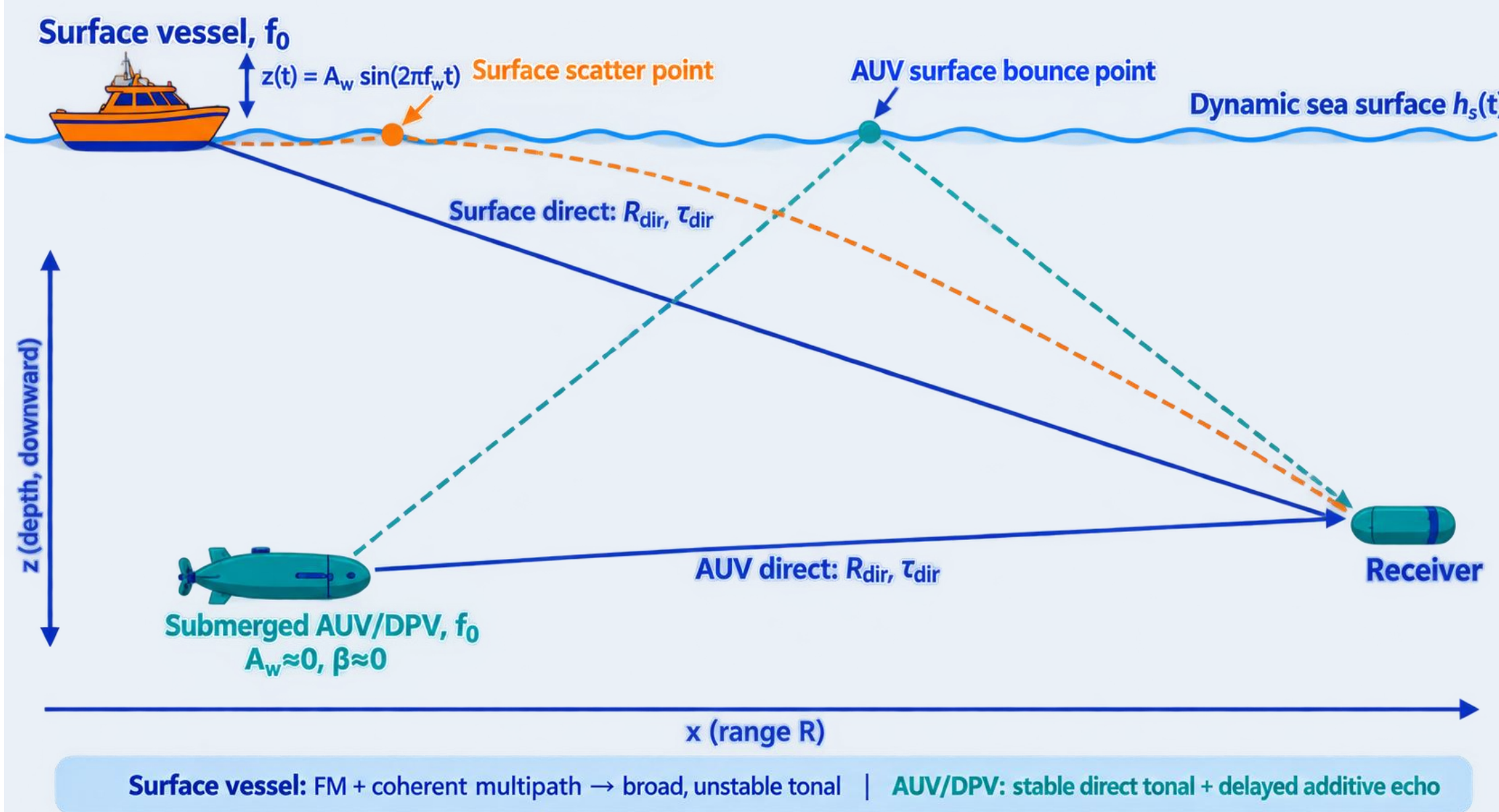
Roy Michael - Department of Marine Technologies, Haifa University

Background

Passive Acoustic Monitoring can distinguish **surface** and **submerged** sources using the effect of wave coupling on mechanical tonal.



Physical Model



Surface: overlapping paths cause fading and Doppler spread. **Submerged:** the surface contribution is a delayed echo.

1. Surface-source geometry

$$\theta_g = \arctan\left(\frac{H_{rec} - H_{src}}{R}\right) \approx 0, \quad \Delta\tau \approx 0$$

2. Heave-driven FM source

$$z(t) = A_w \sin(2\pi f_w t)$$
$$p_{tx}(t) = P_0 \cos(2\pi f_0 t + \beta \sin(2\pi f_w t))$$
$$\beta = \frac{2\pi f_0}{c} A_w \cos \theta_g$$

The resulting tonal is broader and less stable.

3. Underwater two-path channel

$$p_{rx}(t) = \frac{1}{R_{dir}} p_{tx}(t - \tau_{dir}) + \frac{V(t)}{R_{surf}} p_{tx}(t - \tau_{surf})$$
$$V(t) = -R_0 \exp\left(i \frac{4\pi f_0}{c} h_s(t) \sin \theta_g\right)$$

4. Surface-source result

$$p_{rx}(t) \propto p_{tx}(t - \tau_{dir}) \left[1 - R_0 \exp\left(i \frac{4\pi f_0}{c} h_s(t) \sin \theta_g\right) \right]$$

FM carrier $\times$ dynamic multipath envelope $\rightarrow$ bandwidth expansion, fading, and reduced persistence.

5. Underwater-source result

$$A_w \approx 0 \Rightarrow \beta \approx 0, \quad \Delta\tau > 0$$

Stable direct tonal + delayed additive surface echo.

The model predicts broader, less persistent energy for the surface motorboat and narrow, stable tonal bands for submerged DPVs.

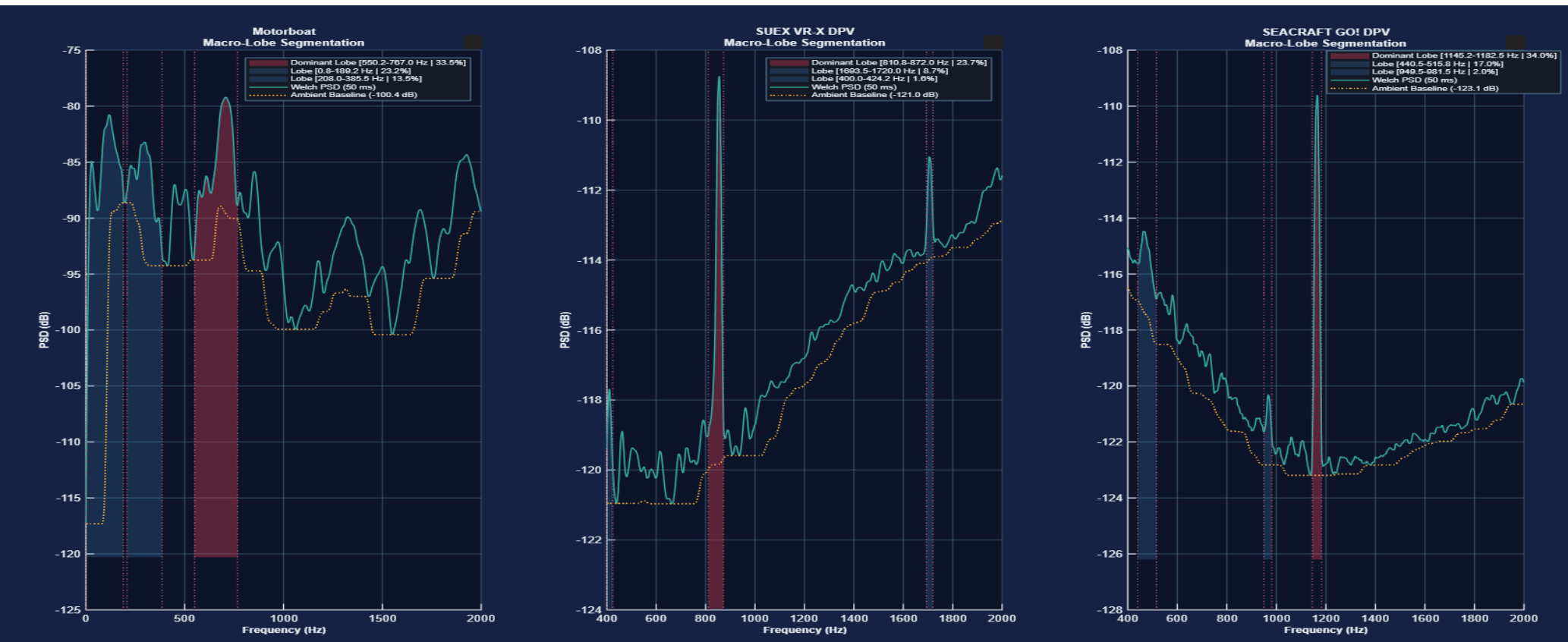
Methodology



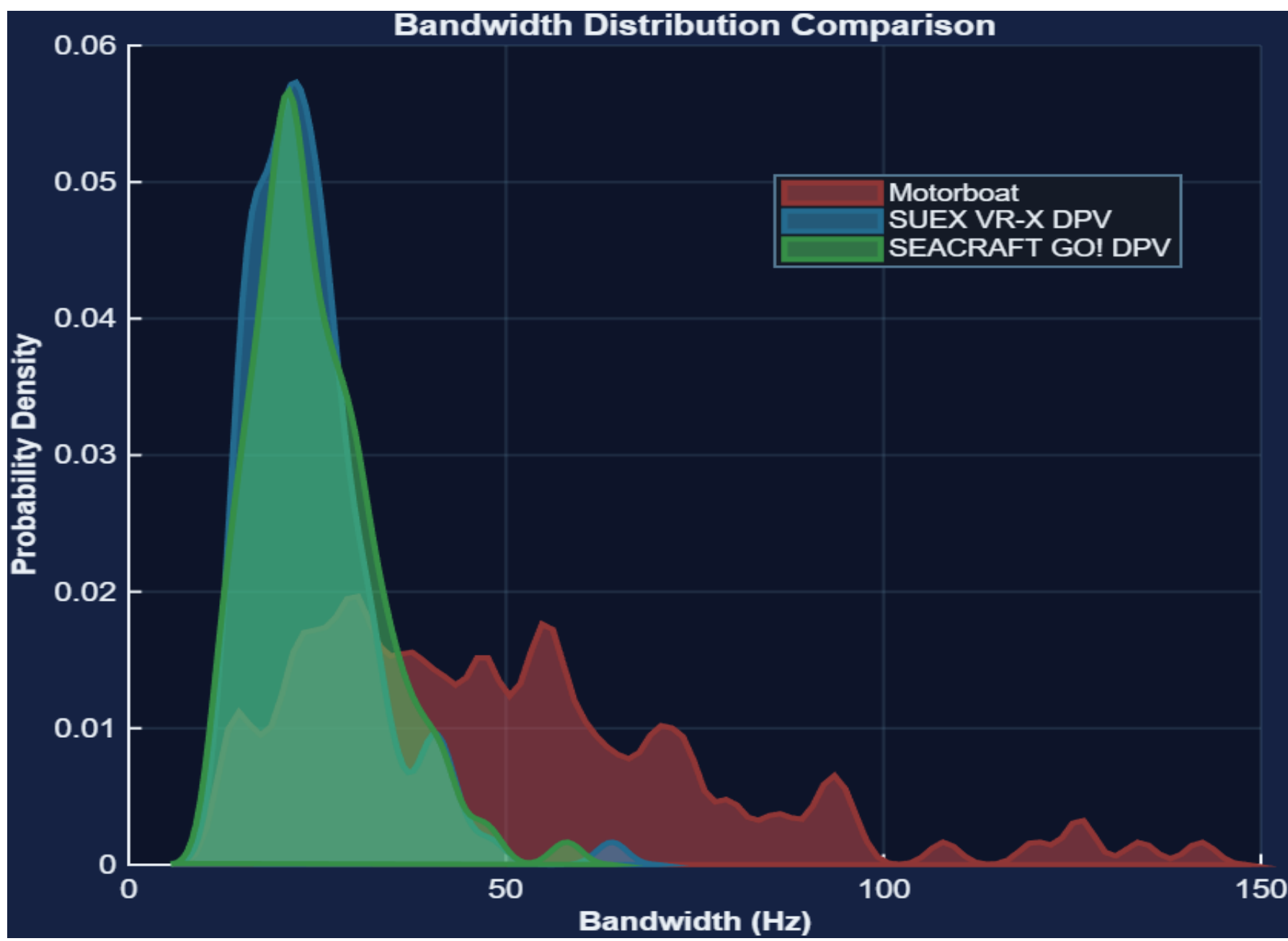
Results

Spectral Fingerprints of Vessel Depth

Dominant spectral peaks



Bandwidth Distribution

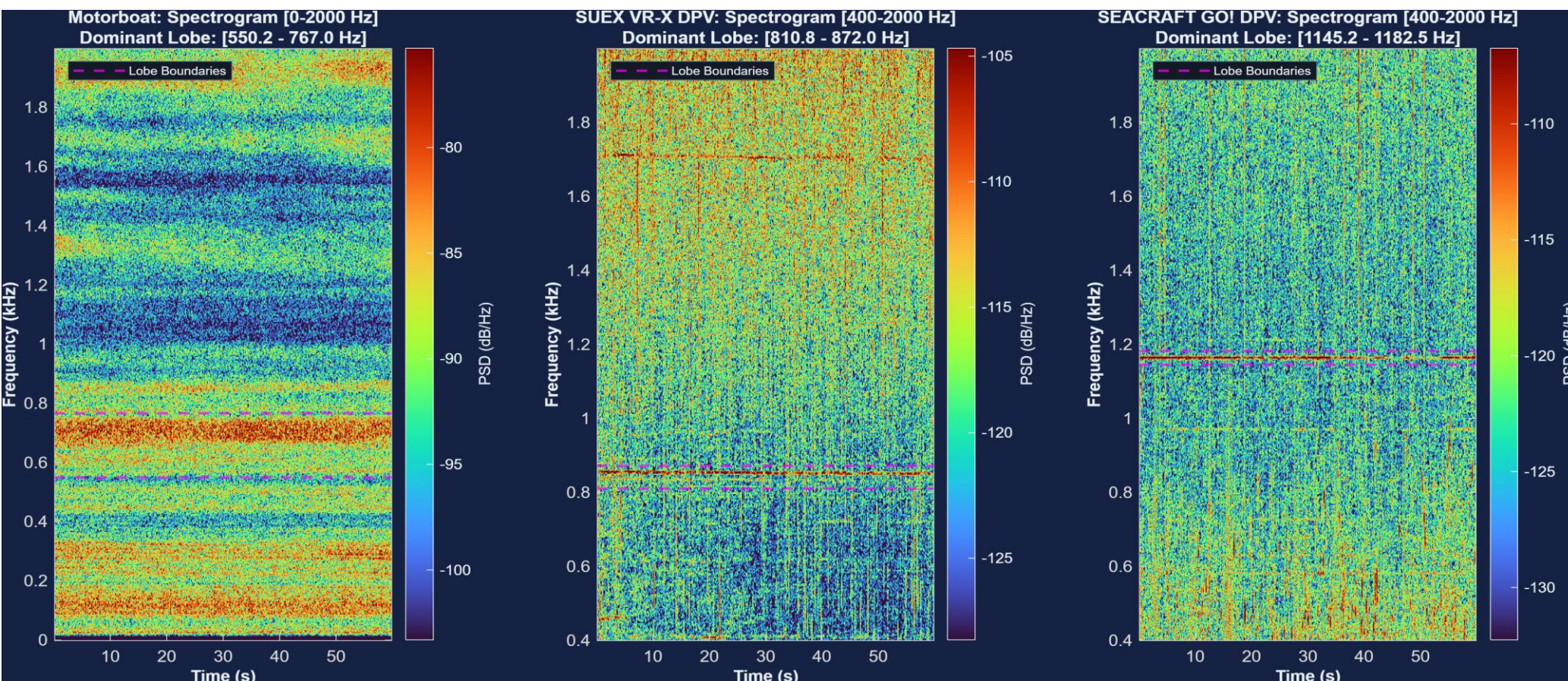


Bandwidth separates source depth

- **Surface motorboat:** 136 Hz
- **Submerged DPVs:** 18–26 Hz

Broad spectral distribution vs. narrow tonal bands

Spectrogram Structure

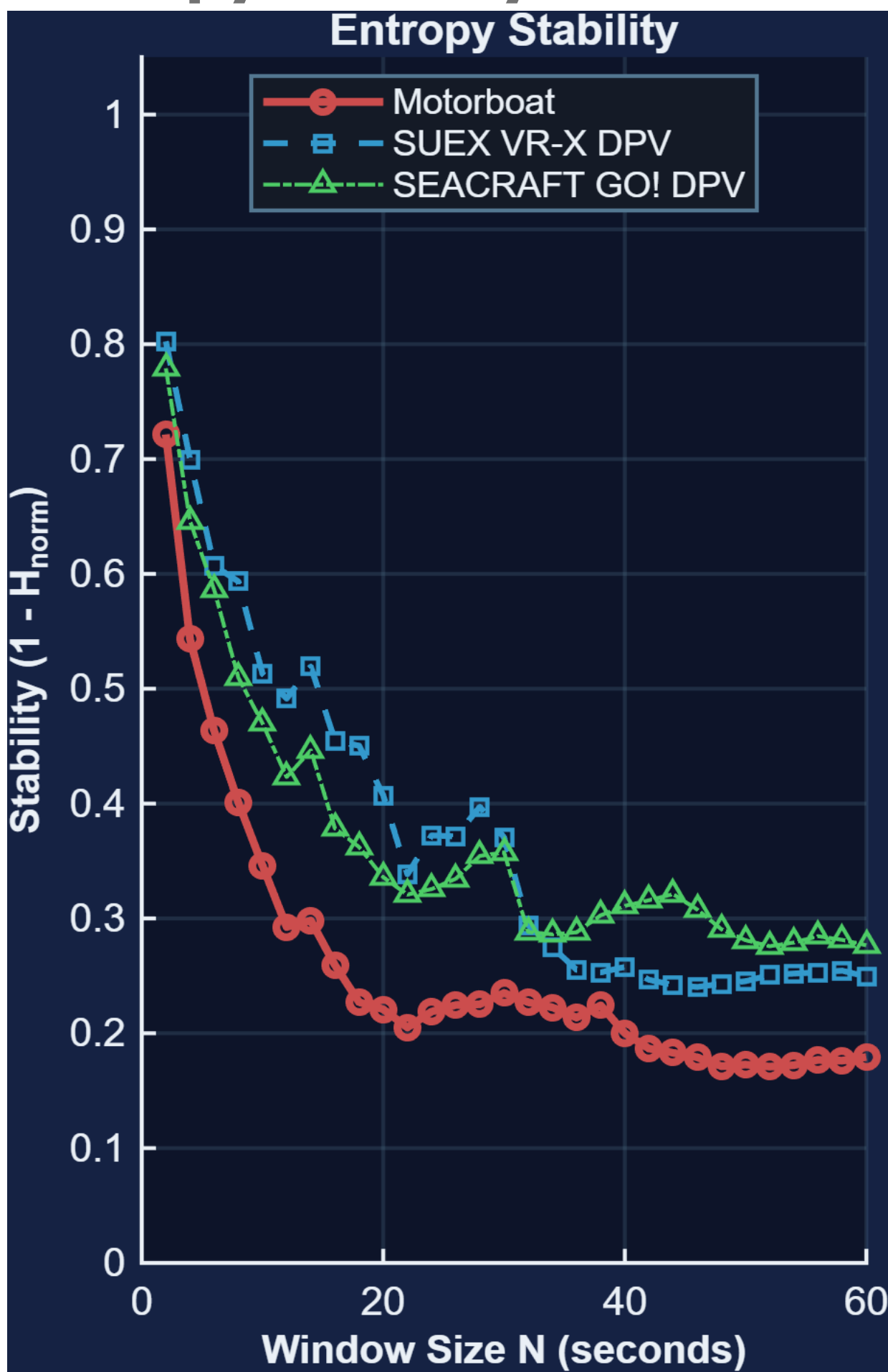


Temporal Stability Separates Depth

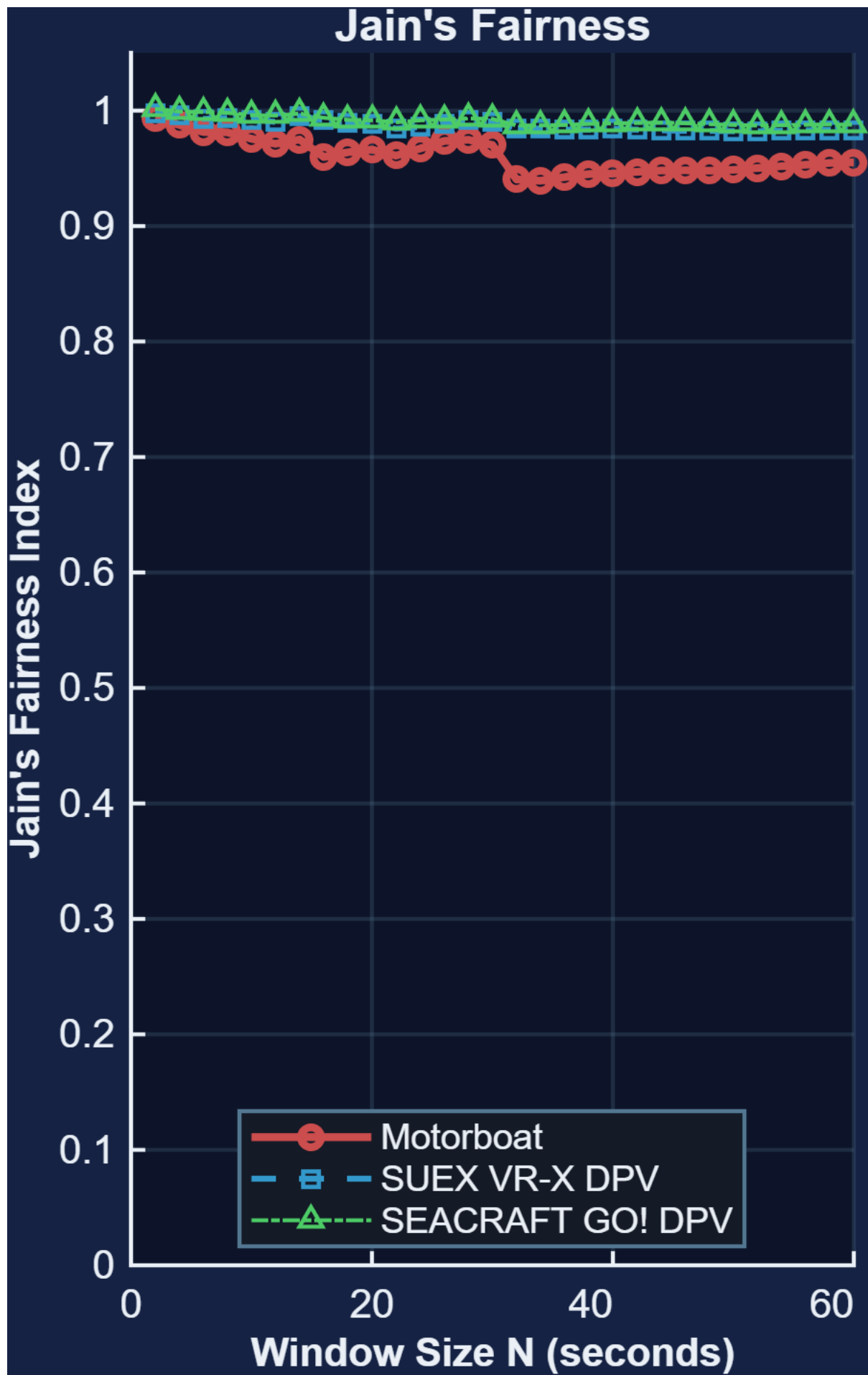
Time-in-Band



Entropy stability



Jain's fairness



Submerged DPV tonal bands persist across longer windows; Surface motorboat stability decreases.

Conclusion and future work

Takeaway

Spectral bandwidth and temporal persistence provide complementary, physics-informed cues for distinguishing surface and submerged vessels.

Next steps

Validate across more vessel types, operating conditions, ranges, and environments; then integrate the features into an ML classifier and real-time detector.

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BREAKING THE SURFACE

